

Optimizitation and Validation of Okafor Reduced Reaction Mechanism

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Traditional sources of energy rely on combustion of fossil fuels that produce carbon emissions which have adverse effects on the environment:

- Global Warming
- Climate Change

New carbon-free alternative fuels are needed to prevent exacerbating the effects of carbon pollutants

- The Japanese government seeks to reduce total greenhouse gas emissions by 46% before 2030
- By 2050, Japan envisions its cities to be carbon-neutral

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Ammonia, NH_3 , is gaining interest among researchers for its potential to meet energy needs in a variety of applications.

Ammonia is a suitable fuel for combustion application because

- Ammonia combustion produces no CO_2
- Ammonia has a high energy density — both gravimetric and volumetry energy densities
- Easier to store than hydrogen
- Infrastructure to produce and transport ammonia already exists

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Drawbacks of using ammonia as a combustion fuel:

- Laminar burning velocity of ammonia is slower than methane (CH_4) and hydrogen (H_2)
- Ammonia has a narrow flammability range
- High NO_x emissions

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Abe Mechanism is developed based on Okafor Reduced Reaction Mechanism¹ to improve laminar burning velocity predictions for $\text{NH}_3/\text{H}_2/\text{Air}$ flames.

- Introduces N_2H_x reaction pathways for more detailed $\text{NH}_2 \rightarrow \text{NH}$ reactions
- Recognises the importance of $\text{NNH} \leftrightarrow \text{N}_2 + \text{H}$ in reaction zone of the flame
- Updated constants for Arrhenius function for reactions

¹[Ekenechukwu Chijioke Okafor et al.](#) “Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism”. In: *Combustion and Flame* 204 (June 1, 2019), pp. 162–175. ISSN: 0010-2180. DOI: [10.1016/j.combustflame.2019.03.008](https://doi.org/10.1016/j.combustflame.2019.03.008).

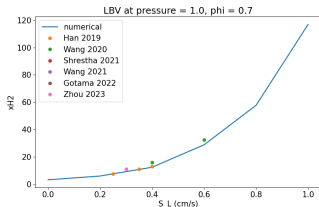
- Numerically simulate flames with Abe mechanism at pressures 1, 3 and 5 atm for lean, stoichiometric and rich equivalence ratios.
 - Flame 1: $\text{NH}_3/\text{H}_2/\text{Air}$
 - Flame 2: $\text{CH}_4/\text{NH}_3/\text{Air}$
- Properties evaluated:
 - Laminar burning velocity
 - Species profiles of NO , NH_3 , CO_2
 - Intermediate species profiles of OH , NNH , NH
- Validate by comparing numerically simulated values to experimental data from literature
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

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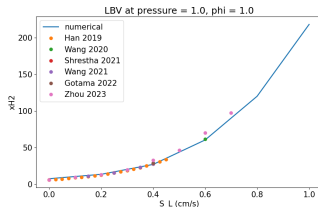
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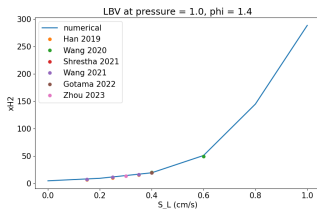
Laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flame at 1atm of pressure



(a) lean equivalence ratio

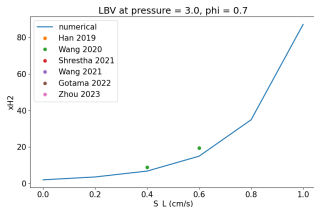


(b) stoichiometric equivalence ratio

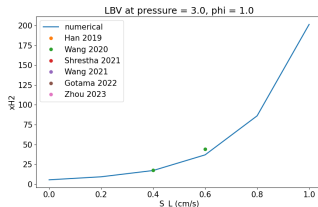


(c) rich equivalence ratio

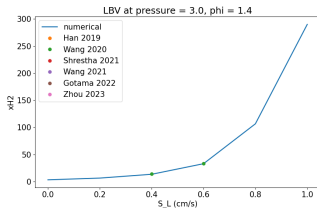
Laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flame at 3atm of pressure



(a) lean equivalence ratio

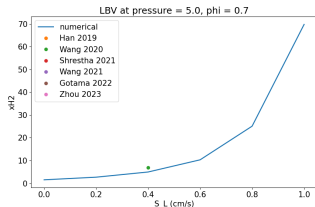


(b) stoichiometric equivalence ratio

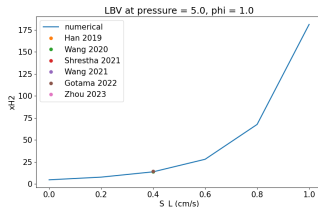


(c) rich equivalence ratio

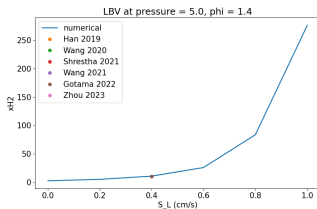
Laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flame at 5atm of pressure



(a) lean equivalence ratio

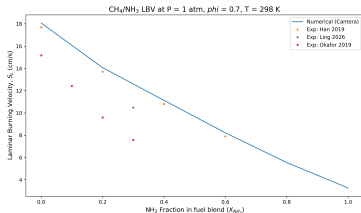


(b) stoichiometric equivalence ratio

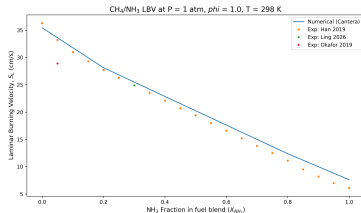


(c) rich equivalence ratio

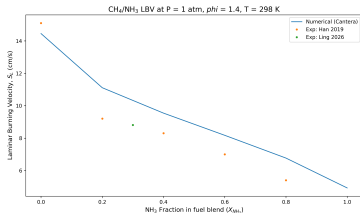
Laminar burning velocity of $\text{CH}_4/\text{NH}_3/\text{Air}$ flame at 1atm of pressure



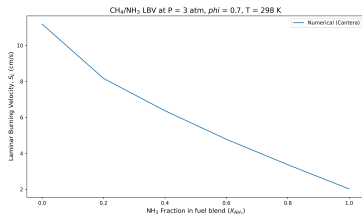
(a) lean equivalence ratio



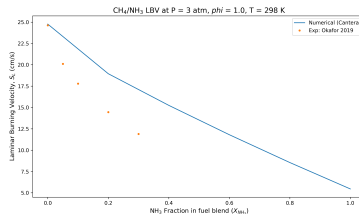
(b) stoichiometric equivalence ratio



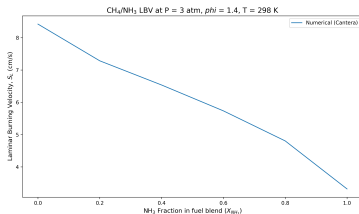
Laminar burning velocity of $\text{CH}_4/\text{NH}_3/\text{Air}$ flame at 3atm of pressure



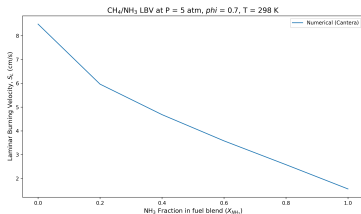
(a) lean equivalence ratio



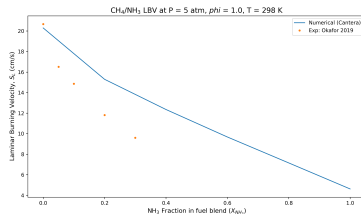
(b) stoichiometric equivalence ratio



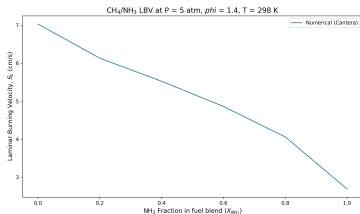
Laminar burning velocity of $\text{CH}_4/\text{NH}_3/\text{Air}$ flame at 5atm of pressure



(a) lean equivalence ratio



(b) stoichiometric equivalence ratio



- Numerically evaluate species profiles with Plug Flow Reactor
 - NO
 - NH₃
 - CO₂
- Numerically evaluate intermediate species profiles with Stagnation Flame Model
 - OH
 - NNH
 - NH
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

- Gabriel J. Gotama et al. "Measurement of the Laminar Burning Velocity and Kinetics Study of the Importance of the Hydrogen Recovery Mechanism of Ammonia/Hydrogen/Air Premixed Flames". In: *Combustion and Flame* 236 (Feb. 1, 2022), p. 111753. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2021.111753. URL: <https://www.sciencedirect.com/science/article/pii/S001021802100496X> (visited on 05/19/2026)
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